A second-order obstruction to localized edge-mode continuation in a non-Hermitian interacting Kitaev chain

Xiaojian Zeng
PBC School of Finance, Tsinghua University, 100190
xiaojianzeng@foxmail.com

Abstract

An edge commutator zero mode on an exactly solvable slice need not admit a smooth localized continuation away from that slice. We establish a controlled example in the dimerized non-Hermitian interacting Kitaev chain with its physical open left boundary. At $U/t = \delta/t = \eta = 1/5$, $\mu = 0$, and $\Delta/t = 1 + \epsilon$, the normalized edge operator at $\epsilon = 0$ has a unique parity-odd exponentially localized first-order correction. A five-mode continuum resonance nevertheless obstructs every exponentially localized second-order correction. The argument combines a bounded half-line spectral similarity, uniqueness across all Clifford degrees, and a nonzero on-shell forcing functional. An exact dyadic finite witness, checked with ball arithmetic and a weighted inverse estimate, gives $0.0001054530 < \mathrm{Re}\,\Phi/t < 0.0001525309$ for the infinite half-line forcing in the specified scattering normalization. The error includes the physical bond crossing the witness cutoff and the complete omitted source tail. A separately generated witness also excludes zero. The obstruction persists despite positive one-particle real-line gaps and a regular first-order correction. It excludes the stated twice-differentiable local continuation; it does not determine a finite-perturbation phase boundary or exceptional point, or exclude nonanalytic branches and branches with nonlocal derivatives.

1 Introduction

Exact strong zero modes in interacting chains provide a stringent form of boundary-operator structure. Fendley's construction for the uniform XYZ chain shows that such structure can survive interactions [1]. Its survival under changes of the Hamiltonian, however, depends on more than the existence of a localized mode in an unperturbed quadratic representation. Perturbative resonances and overlapping excitation bands already play a central role in the stability of boundary strong zero modes in other models [2]. A vanishing perturbative denominator alone is insufficient to decide continuation: the forcing can vanish on the resonant channel, and homogeneous solutions can change the next-order equation.

We study the non-Hermitian interacting Kitaev model of Sayyad and Lado [3]. Their exactly solvable slice has equal hopping and pairing, zero chemical potential, and common dimerization of the hopping, pairing, and complex interaction. We vary the pairing away from that slice at a specified non-Hermitian, dimerized point. The result is a second-order obstruction for a normalized edge *commutator* operator on the original half-line. We do not infer a many-body phase boundary from the operator equation.

The decisive step is a nonzero forcing functional at a real-momentum $3A - 2B$ resonance, where A and B denote the two quadratic Majorana subchains. We first construct the localized first-order correction and prove its uniqueness among all parity-odd localized operators. We then bound the forcing error for the exact infinite solution using a finite dyadic witness. This removes both the finite-ansatz ambiguity and the uncontrolled thermodynamic extrapolation that would otherwise limit a resonance test. The scope is model-specific. Uniform XYZ constructions and the recent commutant-algebra study of strong zero modes [4] are prior work, as is the general resonance mechanism [2].

2 Model, boundary and continuation problem

For open finite chains, write the defining fermionic Hamiltonian as

$$H_L = -\sum_{j=1}^{L-1} m_j \left[t(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j) + \Delta(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j)\right] + (U - \mathrm{i}\delta)\sum_{j=1}^{L-1} m_j(2n_j - 1)(2n_{j+1} - 1) - \mu\sum_{j=1}^{L}(n_j - \tfrac{1}{2}), \qquad m_j = 1 + (-1)^{j+1}\eta. \tag{1}$$

Here $\{c_j, c_k^\dagger\} = \delta_{jk}$, $n_j = c_j^\dagger c_j$, and $t > 0$ sets the energy unit. All four bond parameters have the same dimerization. The leftmost bond is $1 + \eta$, with no boundary modification. Equation (1) is the model in Eqs. (1)–(2) of Ref. [3]. We set $t = 1$ in the derivation and restore it in dimensional statements. Thus $\epsilon = \Delta/t - 1$, all operators Q_j, and the cell momenta are dimensionless, whereas frequencies, commutators and Φ have energy units. This is a static effective non-Hermitian Hamiltonian; no dissipative master equation is specified.

At $U = \delta = \eta = 1/5$ and $\mu = 0$, the Jordan–Wigner spin representation is

$$H(\epsilon) = H_0 + \epsilon V, \tag{2}$$

$$H_0 = \sum_{j\geq 1} m_j[-X_j X_{j+1} + g Z_j Z_{j+1}], \qquad g = \frac{1 - \mathrm{i}}{5}, \tag{3}$$

$$V = \frac{1}{2}\sum_{j\geq 1} m_j[-X_j X_{j+1} + Y_j Y_{j+1}]. \tag{4}$$

X_j, Y_j, Z_j are Pauli matrices on the physical site j. The infinite expressions mean local derivations on the half-line spin algebra, rather than bounded infinite Hamiltonians. We use the notation $[H, Q]$ for the norm-convergent commutator series on its domain.

An operator Q is exponentially localized at the left end if there are $C_Q, \kappa_Q > 0$ and operators $Q^{[r]}$ supported on sites $1, \ldots, r$ such that

$$\left\| Q - Q^{[r]} \right\| \leq C_Q e^{-\kappa_Q r}. \tag{5}$$

This is locality in the original-spin operator norm. Constants may differ between coefficients. Physical fermion parity is the automorphism induced by $\prod_j Z_j$ on finite local operators.

Define Majoranas

$$\alpha_j = \left(\prod_{k<j} Y_k\right) X_j, \qquad \beta_j = -\left(\prod_{k<j} Y_k\right) Z_j, \qquad \{\gamma_j, \gamma_k\} = 2\delta_{jk}\mathbf{1}, \tag{6}$$

where the last relation uses a single enumeration of both families. Put $A_j = \alpha_j$ for odd j and $A_j = \beta_j$ for even j; let $B_j = \beta_j$ for odd j and $B_j = \alpha_j$ for even j. Every A_j is parity odd and every B_j parity even. Set

$$\rho = \frac{3(1 - \mathrm{i})}{10}, \qquad C = \sqrt{1 - \rho^2}, \qquad Q_0 = C\sum_{n\geq 0} \rho^n \alpha_{2n+1}, \tag{7}$$

using the principal square root. Since $|\rho|^2 = 9/50 < 1$, the series converges in operator norm, $Q_0^2 = \mathbf{1}$, and the half-line recurrence gives $[H_0, Q_0] = 0$. No Hermiticity of Q_0 is required.

Proposition 1. *For the physical half-line Hamiltonian (3)–(4), there is a unique parity-odd exponentially localized Q_1 satisfying*

$$[H_0, Q_1] = -[V, Q_0], \qquad \{Q_0, Q_1\} = 0. \tag{8}$$

There is no exponentially localized Q_2 satisfying

$$[H_0, Q_2] = -[V, Q_1]. \tag{9}$$

Consequently, there is no twice norm-differentiable exact commutator-zero-mode branch $Q(\epsilon)$ based at Q_0, with exponentially localized first two Taylor coefficients and square normalization to first order.

For the last statement we write $Q(\epsilon) = Q_0 + \epsilon Q_1 + \epsilon^2 Q_2 + o(\epsilon^2)$ in norm, so $Q_2 = Q''(0)/2$. The branch is understood to be in the commutator domain at each nearby ϵ. Section 6 justifies extracting the coefficient equations without assuming an infinite Hamiltonian is bounded.

3 Half-line spectral representation and uniqueness

3.1 Bounded similarity

Let $a = 6/5$ and $b = 4/5$. Direct Clifford multiplication gives two subchains, each of the form

$$H_s = \mathrm{i} \sum_{n \geq 1} (p_s u_n v_n + h_s v_n u_{n+1}), \qquad (p_A, h_A) = (ga, b), \quad (p_B, h_B) = (a, gb), \tag{10}$$

where $u_n = A_{2n-1}, v_n = A_{2n}$ in A, and similarly for B. On coefficient vectors ordered as (u, v), commutation is

$$K_s = \begin{pmatrix} 0 & 2\mathrm{i}D_s \\ -2\mathrm{i}D_s^T & 0 \end{pmatrix}, \qquad D_s = p_s \mathbf{1} - h_s S, \qquad (Sf)_n = f_{n-1}, \quad f_0 = 0. \tag{11}$$

The superscript T denotes the bilinear transpose, not the Hermitian adjoint. Since $S^T S = \mathbf{1}$,

$$G_s = D_s^T D_s = (p_s^2 + h_s^2)\mathbf{1} - p_s h_s (S + S^T). \tag{12}$$

The real unitary sine transform diagonalizes $S + S^T$, giving the symbol $G_s(q) = p_s^2 + h_s^2 - 2p_s h_s \cos q$, $0 < q < \pi$. Its real part is strictly positive for both subchains at our parameters. Hence $G_s^{1/2}$ and $G_s^{-1/2}$ are bounded normal multiplication operators, with the positive-real-part square-root branch.

Set $J_s = D_s G_s^{-1/2}$. Then $J_s^T J_s = \mathbf{1}$. For B, $|h_B| < |p_B|$ makes D_B invertible by a Neumann series, so $J_B J_B^T = \mathbf{1}$. For A, D_A is bounded below by $|h_A| - |p_A| > 0$ and has closed range of codimension one. Indeed its sole left-null condition is determined by

$$q_n = C\rho^{n-1}, \quad D_A^T q = 0, \quad q^T q = 1, \qquad J_A J_A^T = \mathbf{1} - P_0, \quad P_0 = qq^T. \tag{13}$$

The codimension follows either from the one-step recurrence for $D_A f = y$ or from the index of the unilateral shift. Its range is the bilinear orthogonal complement of q.

The bounded invertible map

$$\mathcal{S}_A(c, f, k) = (qc + J_A f, k), \qquad \mathcal{S}_A^{-1}(u, v) = (q^T u, J_A^T u, v) \tag{14}$$

separates the discrete zero mode. Use $\mathcal{S}_B(f, k) = (J_B f, k)$ for B. On the continuum coordinates the transformed matrix is $\left(\begin{smallmatrix} 0 & 2\mathrm{i}\sqrt{G} \\ -2\mathrm{i}\sqrt{G} & 0 \end{smallmatrix}\right)$. The constant orthonormal eigenvectors $(\mathrm{i}, 1)/\sqrt{2}$ and $(-\mathrm{i}, 1)/\sqrt{2}$ give signed frequencies

$$\pm\omega_s(q), \qquad \omega_s(q) = 2\sqrt{G_s(q)}, \quad \mathrm{Re}\,\omega_s(q) > 0. \tag{15}$$

This is a bounded similarity of the non-Hermitian half-line generator, not an application of the self-adjoint spectral theorem to K_s.

3.2 All-degree localized kernel

Clifford monomials are an orthonormal basis of the tracial L^2 space of the half-line spin algebra. Quadratic commutation preserves each degree; its degree-k action is the additive exterior-power generator of $K_A \oplus K_B$. The preceding similarities lift to bounded invertible maps at every fixed degree. A bound uniform in degree is unnecessary.

Every transformed block is multiplication by a sum of independently variable signed continuum frequencies, with at most one discrete A zero factor. Any sum containing a continuum variable is a nonconstant analytic function of that variable because $g \neq 0$. Its zero set has product measure zero, by one-variable analyticity and Fubini's theorem. Coincident momenta with opposite frequencies do not yield L^2 zero vectors: such vectors would have to be supported on a measure-zero diagonal. The exterior construction therefore introduces no additional point kernel.

The only parity-odd tracial L^2 kernel vector is a multiple of Q_0. An operator satisfying (5) belongs to this L^2 space. Thus the difference of any two parity-odd localized solutions of the first equation in (8) is cQ_0. The normalization equation gives $2c\mathbf{1} = 0$. In particular, arbitrary unforced cubic and higher-degree homogeneous additions have been excluded, rather than set to zero by an ansatz. Without a parity restriction the only additional localized kernel vector is the degree-zero identity. A homogeneous term $c\mathbf{1} + dQ_0$ has anticommutator $2cQ_0 + 2d\mathbf{1}$ with Q_0, so square normalization removes both terms. Thus the continuation consequence does not hide a parity assumption on its first derivative.

4 Existence and control of the first-order correction

The perturbation has quadratic and quartic parts. In the physical-site A-before-B ordering,

$$\begin{aligned} V_2 &= \frac{\mathrm{i}}{2}\left(\sum_{j\ \mathrm{odd}} m_j B_j B_{j+1} + \sum_{j\ \mathrm{even}} m_j A_j A_{j+1}\right), \\ V_4 &= -\frac{1}{2}\sum_{j\geq 1} m_j A_j A_{j+1} B_j B_{j+1}. \end{aligned} \tag{16}$$

Consequently $-[V, Q_0]$ contains only linear A and cubic AB^2 terms. The linear solution follows by differentiating the free edge mode under $t \mapsto t(1+\epsilon/2)$:

$$Q_{1,\mathrm{lin}} = C\sum_{n\geq 0} \rho^n \left[\frac{\rho^2}{2(1-\rho^2)} - \frac{n}{2}\right] \alpha_{2n+1}. \tag{17}$$

It is exponentially localized and normalized by differentiation of $Q_0^2 = \mathbf{1}$. Only the cubic correction can contribute to the second-order A^3B^2 forcing.

Let $f = -[V_4, Q_0]$ and let $K_3 = K_A + K_B^{(1)} + K_B^{(2)}$ act on $A \otimes \bigwedge^2 B$ coefficients. The identity $\{Q_0, [V_4, Q_0]\} = [V_4, Q_0^2] = 0$ gives $P_{0,A} f = 0$. Introduce the auxiliary inverse

$$\widetilde{K} = K_3 + 8P_{0,A}. \tag{18}$$

The number 8 has energy units before setting $t = 1$. This auxiliary projector does not change a physical bond or the normalized solution: its role is to shift the otherwise unbounded inverse in the unforced zero-A block. Since the shift makes that block invertible, $P_{0,A}\widetilde{K}^{-1} f = 0$ and $K_3\widetilde{K}^{-1} f = f$.

The real parts of the positive continuum bands have exact ranges

$$[A_-, A_+] = [28/25, 52/25], \qquad [B_-, B_+] = [52/25, 68/25]. \tag{19}$$

For completeness, put $x = -\cos q$. The radicands have positive real part and $\arg(p_s h_s) = -\pi/4$. Since $d\omega_s/dx = 4p_s h_s/\omega_s$, its real part is positive. Evaluating the endpoints gives (19). Direct quadratic minimization of $|G_s|^2$ also gives

$$\inf|G_A| = \frac{328}{625\sqrt{2}}, \qquad \inf|G_B| = \frac{692}{625}. \tag{20}$$

Every shifted cubic multiplier has modulus at least

$$d = \min\{A_- - (B_+ - B_-),\ 2B_- - A_+,\ 8 - 2B_+\} = \frac{12}{25}. \tag{21}$$

The three terms cover opposite B signs, equal B signs, and the shifted discrete A factor, respectively.

Useful conservative bounds are

$$j_s = \frac{(|p_s| + |h_s|)^2}{\inf|G_s|}, \qquad q_*^2 = \frac{|1-\rho^2|}{1-|\rho|^2},$$
$$\operatorname{cond}(\mathcal{S}_A) \le q_*^2 + j_A, \qquad \operatorname{cond}(\mathcal{S}_B) \le j_B, \qquad \left\|\widetilde{K}^{-1}\right\| \le M = \frac{(q_*^2 + j_A)j_B^2}{d} < 33.318. \tag{22}$$

Here $j_A, j_B > 1$. Bilinear transposition preserves the operator norm, which justifies using the same bounds for the inverse maps.

For cubic coefficients define a diagonal weight W_α by $\exp[\alpha(n_A + n_{B_1} + n_{B_2})]$, where the cell indices start at zero, and choose $\alpha = 1/1000$. Appendix A gives an explicit bound

$$\left\|W_\alpha \widetilde{K} W_\alpha^{-1} - \widetilde{K}\right\| \le E < 0.013647916, \quad ME < 0.454721, \quad \left\|W_\alpha \widetilde{K}^{-1} W_\alpha^{-1}\right\| < 61.103. \tag{23}$$

The conjugated identity is first defined on finite sequences. Its bounded difference from $\widetilde{K}$ and the Neumann bound produce a weighted solution; since W_α^{-1} is bounded, it agrees with the unique unweighted solution. The geometric source belongs to the weighted space. Cauchy–Schwarz with a smaller positive weight converts weighted coefficient ℓ^2 summability at fixed degree into absolute coefficient summability and an exponential operator-norm tail. Therefore $Q_{1,\mathrm{cub}} = \widetilde{K}^{-1} f$ has the locality in Eq. (5). Together with (17) and the all-degree kernel argument, this proves the first assertion of Proposition 1.

5 A nonzero resonant second-order forcing

5.1 Half-line left rows and the exact resonance

For an interior momentum q and either signed frequency ω, define the bilinear left scattering row

$$\ell_{v,n} = \sin(nq), \qquad \ell_{u,n} = -\frac{2\mathrm{i}}{\omega}[p\sin(nq) - h\sin((n-1)q)]. \tag{24}$$

Both column recurrences give $\ell K = \omega \ell$. At the open end $\sin(0q) = 0$ enforces the original boundary. The rows are bounded and, in A, annihilate the discrete mode since $p - h\rho = 0$.

Use three positive A rows and two negative B rows at momenta specified by $x = -\cos q$:

$$x_{A,1} = -99/100, \quad x_{A,2} = -95/100, \quad x_{A,3} = x, \qquad x_{B,1} = y - 1/100, \quad x_{B,2} = y + 1/100. \tag{25}$$

There is an exact root of

$$\Omega(x,y) = \sum_{r=1}^{3} \omega_A(q_{A,r}) - \sum_{s=1}^{2} \omega_B(q_{B,s}) = 0 \tag{26}$$

in the real two-coordinate box of radius 10^{-20} centered near $(0.914006, -0.769582)$. The full exact decimal center and a contraction certificate are given in Appendix B. The root is established by a self-map contraction, not by a small residual or by an interval that merely contains zero. All five momenta are interior, and all radicands retain positive real part throughout the box.

For increasing physical A indices $i_1 < i_2 < i_3$ and B indices $j_1 < j_2$, define ℓ_5 on $A_{i_1}A_{i_2}A_{i_3}B_{j_1}B_{j_2}$ as

$$\ell_5(A_{i_1}A_{i_2}A_{i_3}B_{j_1}B_{j_2}) = \det[\ell_r^A(i_s)]_{r,s=1}^3 \det[\ell_r^B(j_s)]_{r,s=1}^2. \tag{27}$$

The B rows use the negative frequencies in (24). This fixes the normalization of the forcing functional

$$\Phi(X) = \ell_5([V_4, X]_{A^3B^2}), \qquad X \in AB^2. \tag{28}$$

Its complex value depends on the row normalization. Its nonvanishing is unchanged by any nonzero rescaling of those rows. It is not a decay rate.

5.2 A posteriori bound for the infinite solution

Let X be a finite cubic coefficient vector extended by zero onto the physical half-line. Put $R = f - \widetilde{K}X$. The inverse estimate and a weighted dual bound on Φ give

$$|\Phi(Q_{1,\mathrm{cub}}) - \Phi(X)| \le 1248632935 \, \|W_\alpha R\|_2 \, . \tag{29}$$

The constant is dimensionless; dividing both sides and the residual by t restores the same inequality at general energy units. To obtain it, each of the two B indices of a cubic monomial can overlap at most two quartic bonds. If

$$m_A = \max\{1, 2(|p_A| + |h_A|)/A_-\}, \quad m_B = \max\{1, 2(|p_B| + |h_B|)/B_-\}, \tag{30}$$

the determinant estimates $6m_A^3$ and $2m_B^2$ imply

$$\left\|\Phi W_\alpha^{-1}\right\| \le (4a)(6m_A^3)(2m_B^2)\frac{2}{(1-e^{-2\alpha})^{3/2}} < 20435046. \tag{31}$$

There are two coordinates per cell in each species; the exterior B^2 sum is at most half the full tensor sum. The product of the unrounded weighted inverse and dual bounds is less than 1248632935. The rounded bounds displayed separately in (23) and (31) are not used to take that product.

The primary witness has $L = 96$ physical sites in its finite generator: 95 A coordinates and 96 B coordinates. These are support cutoffs, not modified half-line equations. Its stored binary floating-point coefficients are interpreted as exact dyadic rationals. The certificate reconstructs the quadratic action with Gaussian integers and rational couplings, then encloses the geometric source and C with 224-bit Arb ball arithmetic [5]. It does not trust the finite solver's eigenvectors, residual estimates, or convergence claim.

The residual includes the physical A link across the cutoff, the B link across the cutoff, the normalization term $8P_{0,A}X$, and the entire omitted geometric source. For $N = L/2$, the latter has the conservative bound

$$T_N = \frac{4a|C|e^{3\alpha(N+2)}|\rho|^N}{1 - |\rho|e^{3\alpha}}. \tag{32}$$

It overcounts the two bonds touching each remaining edge coefficient and starts early enough to include every unrepresented source entry. Explicit finite and normalization bounds are in Appendix C.

The finite functional is enclosed over the *whole root box* and is approximately

$$\frac{\Phi(X)}{t} = 0.00012899190678575 7 + 0.00000985287874885 62\,\mathrm{i}. \tag{33}$$

Table 1: Certified error budget for the 96-site dyadic witness. Residuals and forcing errors are in units of t; all displayed upper bounds are rounded outwards. The finite term uses the original physical cutoff links.

Quantity	Upper bound
Weighted finite physical residual	1.111107×10^{-15}
Omitted geometric source	1.309565×10^{-17}
Auxiliary normalization term	1.772754×10^{-14}
Total $\|W_\alpha R\|_2$	1.885174×10^{-14}
Error in Φ from Eq. (29)	2.353890×10^{-5}

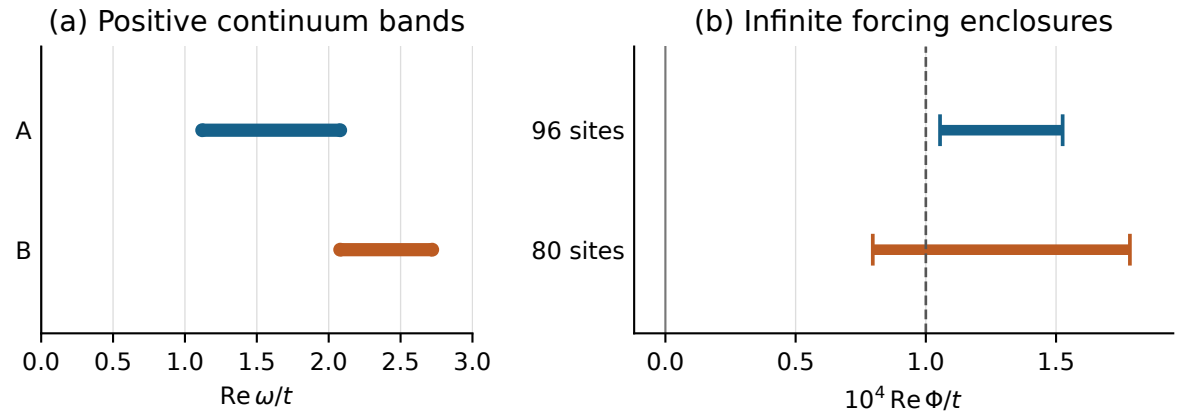


Figure 1: Positive real parts of the one-particle continuum frequencies (left) and certified enclosures of the same infinite on-shell forcing (right). The intervals on the right include the inverse and infinite-tail error; they are not statistical error bars. Both exclude zero, while only the 96-site witness proves $\mathrm{Re}\,\Phi/t > 10^{-4}$ (dashed line). The half-line result is obtained from each complete error bound, not from the apparent agreement of two finite approximations.

Combining its interval enclosure with the unrounded residual bounds gives

$$\boxed{0.0001054530 < \frac{\mathrm{Re}\,\Phi(Q_{1,\mathrm{cub}})}{t} < 0.0001525309.} \tag{34}$$

In particular, the exact infinite functional has real part greater than $0.0001t$. No extrapolation in L enters this inequality.

A second implementation generated an $L = 80$ witness from zero using nearest-neighbour GMRES with residual corrections, without using the first witness as a solver input. Its separately evaluated certificate gives

$$0.0000796284 < \mathrm{Re}\,\Phi/t < 0.0001783554. \tag{35}$$

This independently generated witness excludes zero, but does not by itself establish the stronger 0.0001 threshold. Figure 1 shows the distinct information in the two enclosures.

The algebra and the boundary restoration admit separate checks. Exact occupation-bit and Pauli matrices agree for $L = 2, 4, 6$, including unequal pairing and a nonzero chemical-potential control. The quartic source and the physical parity agree in the original spin algebra. A deterministic Gaussian-integer cubic test gives exact agreement of 156 quintic coefficients between direct Clifford multiplication and the determinant contraction. Omitting the cutoff-crossing bond changes that test functional by a nonzero amount of order $0.005t$, so it detects the relevant boundary error. The small $L = 16$ witness is inconclusive for the infinite functional; its failure to exclude zero is not interpreted as evidence against the larger certificate.

6 Contradiction, operator domain and physical interpretation

If an exponentially localized Q_2 satisfied (9), project that equation onto A^3B^2. The unique Q_1 has only the linear and cubic parts already constructed, and the linear part cannot produce this degree-five forcing. Thus the projected right-hand side is precisely $-[V_4, Q_{1,\mathrm{cub}}]_{A^3B^2}$.

For an exponentially localized operator, its fixed-degree coefficient tails are absolutely summable. Indeed, the number of degree-five monomials with largest physical index r grows only polynomially, whereas tracial L^2 tails are bounded by the exponentially small operator-norm approximation error. Cauchy–Schwarz gives summable ℓ^1 shells. The bounded left rows therefore define an absolutely convergent functional, and finite-range summation by parts has no boundary term at infinity. At the exact root of (26),

$$\ell_5([H_0, Q_2]_{A^3B^2}) = \Omega\ell_5((Q_2)_{A^3B^2}) = 0. \tag{36}$$

The right-hand side of (9), however, has a nonzero functional by (34). This contradiction proves the second assertion of Proposition 1.

There is no hidden bounded-Hamiltonian assumption. If h_j is a uniformly bounded nearest-neighbour interaction and Q satisfies (5), then, for large j, $[h_j, Q^{[j-1]}] = 0$ and $\|[h_j, Q]\| \le 2\,\|h_j\|\, C_Q e^{-\kappa_Q(j-1)}$. The commutator series converges in norm. For a twice norm-differentiable exact branch, pair its equation with an arbitrary local test operator T under the tracial bilinear pairing. Cyclicity gives

$$\tau\big(T[H(\epsilon), Q(\epsilon)]\big) = -\tau\big([H(\epsilon), T]Q(\epsilon)\big). \tag{37}$$

The commutator on T is a finite local sum, so the right-hand side is a bounded functional of $Q(\epsilon)$ and can be differentiated twice. The resulting coefficient equations hold against all local tests. The norm convergence of the coefficient commutators then yields Eqs. (8) and (9) as operator equations. Differentiating the square normalization gives $\{Q_0, Q_1\} = 0$.

The physical conclusion is a controlled failure of smooth, localized edge-operator continuation even when each positive one-particle band is separated from the imaginary axis and the first-order equation is regular. The obstruction is a forced continuum channel at second order. This does not assert that every finite-ϵ edge mode is absent. Nonanalytic branches, branches with nonlocal derivatives, weak edge modes and prethermal operators remain outside the statement. No lifetime, exceptional point, finite-ϵ phase boundary, or many-body eigenvalue-pairing theorem follows. In a non-normal problem, a small commutator alone would not supply spectral conditioning or a conserved normalized expectation value.

The comparison with uniform XYZ is also restricted. Bond alternation here cannot be removed by a global spin-axis rotation or a common energy rescaling. Fendley's uniform construction [1] is not contradicted; the discussion of spatially varying couplings there gives low-order evidence, rather than an all-order result for the present point. The finite boundary modification distinguished in the XYZ treatment of Ref. [4], Sec. VI, is likewise different from the physical half-line used here. Resonance-based boundary obstructions in Ref. [2] motivate the mechanism, but their Hermitian interface models do not provide this complex on-shell forcing and its tail control. These comparisons support a concrete restricted contribution, without an exhaustive priority claim. The broader problem of finding an exact phase boundary or exceptional-point condition away from the solvable Kitaev slice remains unresolved.

Data and code availability

The accompanying reproducibility files contain the two dyadic witnesses, the independent certificate implementation, an isolated replay driver, reference output records, and the script and data for Fig. 1. The README specifies the environment and commands. The numerical bounds support the analytic half-line argument given here; finite computations alone do not prove spectral completeness or locality.

Acknowledgements and use of generative AI

OpenAI Codex was used for literature assistance, formulation and analysis of the restricted claim, symbolic and numerical code, internal verification, and manuscript drafting. GPT-6 was used for manuscript preparation; the complete model/version inventory for earlier phases remains to be confirmed. The checks described here include separate AI-assisted implementations; they are not external peer review. Human scientific review, verification of every cited source, and critical revision of the generated text remain pending for this local draft. The author must confirm the complete tool/version disclosure and take responsibility for the final work before any submission.

A Exponential-weight inverse estimate

With $r = |\rho|^2$, let $W_s q$ have components $e^{s(n-1)} q_n$. Geometric sums give

$$\begin{aligned} N(s)^2 &= \|W_s q\|_2^2 = \frac{|C|^2}{1 - re^{2s}}, \\ D(s)^2 &= \|(W_s - \mathbf{1})q\|_2^2 = |C|^2 \left[\frac{1}{1 - re^{2s}} - \frac{2}{1 - re^{s}} + \frac{1}{1 - r} \right]. \end{aligned} \tag{38}$$

Conjugating the three nearest-neighbour factors changes their sum by at most $2(|h_A|+2|h_B|)(e^{\alpha}-1)$. For the bilinear rank-one projector,

$$\left\| W_\alpha P_0 W_\alpha^{-1} - P_0 \right\| \le D(\alpha)N(-\alpha) + N(0)D(-\alpha). \tag{39}$$

Hence a valid value of E in (23) is

$$E = 2(|h_A| + 2|h_B|)(e^{\alpha} - 1) + 8[D(\alpha)N(-\alpha) + N(0)D(-\alpha)]. \tag{40}$$

Using (22), the weighted inverse is bounded by $M/(1 - ME)$. The enclosed values are $M < 33.317951$, $ME < 0.454720583$, and $M/(1 - ME) < 61.102528$. These bounds concern a fixed Clifford degree and a specified exponential weight, with no all-degree norm estimate assumed.

B Enclosure of the two-coordinate resonance

The exact decimal center is

$$\begin{aligned} x_c &= \texttt{0.914005998943111754560338502271033241599810017455941304810241}, \\ y_c &= \texttt{-0.769581730425783703625420540788997375179263443033756807855182}. \end{aligned}$$

Let $F = (\operatorname{Re}\Omega, \operatorname{Im}\Omega)$ and B_c be the real infinity-norm box with this center and radius $r_c = 10^{-20}$. The checker constructs an exact dyadic matrix C_0 by rounding the inverse central Jacobian to binary floating point, then interpreting each entry as its exact binary rational. All following evaluations use ball arithmetic. It verifies C_0 is nonsingular and obtains

$$\|C_0 F(x_c, y_c)\|_\infty < 3.257 \times 10^{-61}, \qquad \sup_{B_c} \|\mathbf{1} - C_0 DF\|_\infty < 2.762 \times 10^{-16}. \tag{41}$$

Thus $z \mapsto z - C_0 F(z)$ maps B_c into itself and is a contraction, because the first bound plus r_c times the second is less than r_c. Its fixed point is a zero of F. The differentiations are explicit:

$$\partial_x \Omega = \frac{4 p_A h_A}{\omega_A(x)}, \qquad \partial_y \Omega = -4 p_B h_B \left[\frac{1}{\omega_B(y - 1/100)} + \frac{1}{\omega_B(y + 1/100)} \right], \tag{42}$$

where $\omega_s(x) = 2\sqrt{p_s^2 + h_s^2 + 2p_s h_s x}$ in this appendix. The supplied code records C_0 and evaluates the left rows over the entire box. The approximate decimal center in the main text is not the certificate input.

C Residual and determinant contraction

Use zero-based physical indices in this appendix, so $m_j = 6/5$ for even j and $4/5$ for odd j. Write a cubic vector as $X = \sum_{i,b<c} X_{i,bc} A_i B_b B_c$ and extend $X_{i,cb} = -X_{i,bc}$. For the L-site witness the array shape is $(L-1, L, L)$. It is embedded into $(L, L+1, L+1)$ before applying the nearest-neighbour commutator. This includes every link that acts on its support. After a common power-of-two denominator is extracted from the binary coefficients, the rational quadratic action is performed with unbounded Gaussian integers. The geometric forcing and normalization are then enclosed with balls.

Let r_{fin} be the coefficient ℓ^2 residual of the restored physical action with its explicitly represented source terms, and $Z_{bc} = \sum_n C\rho^n X_{2n,bc}$. Triangle inequalities give

$$\|W_\alpha R\|_2 \le e^{3\alpha(N+1)} r_{\mathrm{fin}} + T_N + 8N(\alpha) e^{2\alpha(N+1)} \|Z\|_2 , \qquad N = L/2. \tag{43}$$

The finite weights deliberately overestimate every represented output index. T_N in (32) covers the entire complement, so changing which of the last geometric source entries is explicitly represented does not change the logical coverage of the bound. The normalization term is necessary even when a numerical solve intended to enforce $P_{0,A} X = 0$.

For clarity, the finite functional can be evaluated without assembling a five-index tensor. Let a_j be the column of the three A rows at site j, and let $b_s(j)$ be the sth B row. With $T_i(j,s) = \sum_r X_{i,jr} b_s(r)$, define $k = j+1$ and obtain directly from Eq. (16)

$$\Phi(X) = \sum_{i=0}^{L-2} \sum_{j=0}^{L-1} m_j \det(a_j, a_k, a_i) \big[b_1(k) T_i(j,2) - b_2(k) T_i(j,1) \\ - b_1(j) T_i(k,2) + b_2(j) T_i(k,1) \big]. \tag{44}$$

The $T_i(k,s)$ terms vanish for $k = L$ by zero extension. The $j = L-1$ summand is retained: it is the quartic bond crossing the finite support. Direct Clifford multiplication checks the sign and ordering in Eq. (44) independently.

The 96-site witness comes from a finite spectral solve with residual correction; the 80-site witness was generated independently by GMRES from zero. Both are checked by the Gaussian-integer and ball-arithmetic route above. New witnesses, which the supplied checker can generate, require recertification: neither solver tolerances nor matching decimal digits replace Eq. (43).

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